FLIGHT CONTROLLER

ARK Pi6X

Raspberry Pi compute module carrier with an STM32H743 flight controller and 2S–6S power input

The USA-built ARK Pi6X puts a Raspberry Pi Compute Module carrier, an STM32H743 flight controller and a 2S–6S battery input with current monitoring on one board, with 30.5 × 30.5 mm mounting for a 4-in-1 ESC. The Pi runs ARK-OS and reaches the flight controller over USB and a dedicated UART.

The ARK Pi6X Flow is the same board with an optical flow sensor and a distance sensor fitted. PX4 ships by default; ArduPilot is supported.

KEY FEATURES

- STM32H743IIK6, 480 MHz, 2 MB flash, 1 MB RAM
- Dual heated TDK InvenSense ICM-42688-P IMUs on separate SPI buses
- Bosch BMP390 barometer and ST IIS2MDC magnetometer
- Raspberry Pi Compute Module 4; Compute Module 5 with reduced I/O
- 2S–6S battery input, TI INA226 voltage and current monitor
- 8 PWM outputs, DShot on outputs 1–6
- Pi Ethernet, micro HDMI, two MIPI CSI cameras, USB-C host
- Pi6X Flow: PixArt PAW3902 flow and Broadcom AFBR-S50LV85D distance

SPECIFICATIONS

PROCESSOR & COMPUTE	
MCU	ST STM32H743IIK6, 480 MHz
Flash / RAM	2 MB / 1 MB
Compute module	Raspberry Pi CM4; CM4 Lite boots from microSD
Flight controller link	USB through the hub, and UART5 to the Pi

ELECTRICAL	
Battery input	5.5–25.2 V (2S–6S)
Input clamp	26 V standoff, 28.9 V breakdown
Current sensing	INA226, 0.5 mΩ shunt
5 V on CAN, GPS, TELEM1	500 mA, shared with the CAN transceiver
PWM outputs	3.3 V logic, 220 Ω series, signal only

ENVIRONMENTAL & MECHANICAL	
Operating temperature	–10 °C to +60 °C; Pi6X Flow 0 °C to +60 °C
Storage temperature	–10 °C to +60 °C
Dimensions	91.5 × 56 mm, without compute module and ESC
Weight	41 g without compute module, Pi microSD card and ESC; 53 g with CM4 and Pi microSD card
ESC mounting	4-in-1 ESC on 30.5 × 30.5 mm



AT A GLANCE	
MCU	STM32H743IIK6
Compute	Raspberry Pi CM4 / CM5
IMUs	2 × ICM-42688-P
Baro / Mag	BMP390 / IIS2MDC
Input	2S–6S, 5.5–25.2 V
Operating temp	–10 to +60 °C
Flow operating	0 to +60 °C
Dimensions	91.5 × 56 mm
Weight	41 g
Hardware rev	Rev 2.0

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed (ARK Pi6X)

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-pi6x/
arkelectron.com/product/ark-pi6x-flow/

DOCS docs.arkelectron.com/products/flight-controller/ark-pi6x-flow

CAD github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARK Pi6X	ARK Pi6X Flow
Optical flow	—	PixArt PAW3902, SPI3
Distance sensor	—	Broadcom AFBR-S50LV85D, SPI6
Operating temperature	−10 °C to +60 °C	0 °C to +60 °C

Same board, connectors and pinout; the ARK Pi6X leaves both sensors unfitted.

POWER RAILS

RAIL	LIMIT	CONNECTORS
VDD_5V_HIPOWER	500 mA, shared with the CAN transceiver	CAN, GPS, TELEM1
VDD_5V_SBUS_RC	250 mA fuse	RC
5.0V	2 A fuse	Pi LED Strip
5.0V	500 mA fuse each	Pi I2C1, Pi UART3, Pi GPIO, Pi Fan
USB VBUS	500 mA each	USB-C, Pi USB (hub ports 2 and 4)
HDMI 5 V	250 mA fuse	Micro HDMI
3.3V_RPI	600 mA, the compute module's 3.3 V output, shared	CAM0, CAM1, Pi Console

VDD_5V_HIPOWER and the USB ports are MIC2026A current-limited switch channels, guaranteed to 500 mA; above that they may limit anywhere in 0.5–1.25 A. Every 5 V rail and the compute module draw from the board's 5 V regulator.

SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
ICM-42688-P	SPI1	nCS, DRDY	6-axis accelerometer / gyroscope
ICM-42688-P	SPI2	nCS, DRDY	6-axis accelerometer / gyroscope
BMP390	I2C4	0x76	Barometric pressure
IIS2MDC	I2C4	0x1E	3-axis magnetometer
INA226	I2C3	0x41	Battery voltage and current
PAW3902 (Flow)	SPI3	nCS, DRDY	Optical flow
AFBR-S50LV85D (Flow)	SPI6	nCS, IRQ	Time-of-flight distance
GPS connector	I2C1	External	Compass on a GPS module

SERIAL PORT MAPPING

UART	CONNECTOR	PX4	ARDUPILOT
USART1	GPS	GPS1, /dev/ttyS0	SERIAL3
USART3	FC DEBUG	Console, /dev/ttyS1	—
UART4	PWM + UART4	TELEM4, /dev/ttyS2	SERIAL4
UART5	None, wired to the Pi	TELEM2, /dev/ttyS3	SERIAL2
USART6	RC	RC, /dev/ttyS4	SERIAL5
UART7	TELEM1 (RTS/CTS)	TELEM1, /dev/ttyS5	SERIAL1

UART5 reaches the Pi's UART4 (/dev/ttyAMA4) on a CM4 and UART3 (/dev/ttyAMA3) on a CM5.

FIRMWARE

FIRMWARE	BUILD TARGET	NOTES
PX4	`ark_pi6x_default`	Ships installed
ArduPilot	`ARK_PI6X`	—

NOTES

- DShot runs on outputs 1–6. Outputs 7–8 are on TIM12, which has no DMA, so they run PWM and OneShot only.
- The compute module's USB reaches the flight controller, USB-C and both USB JST-GH ports through a full-speed (12 Mbps) hub. A cable in the programming micro USB switches the compute module to that port and disconnects the hub until reboot.
- With a Compute Module 5, only CAM1 works and the Pi UART numbering shifts by one; USB 3.0 isn't wired.
- CAN1 is terminated on the board (120 Ω).
- ArduPilot has no drivers for the Pi6X Flow's PAW3902 flow sensor or AFBR-S50LV85D distance sensor.
- Temperature ratings are for the carrier. The Raspberry Pi CM4 and CM5 are rated −20 °C to +85 °C.

CONNECTOR PINOUTS

PWM + UART4 · 11-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	FMU_CH1	3.3 V
2	FMU_CH2	3.3 V
3	FMU_CH3	3.3 V
4	FMU_CH4	3.3 V
5	FMU_CH5	3.3 V
6	FMU_CH6	3.3 V
7	FMU_CH7	3.3 V
8	FMU_CH8	3.3 V
9	UART4_TX	3.3 V
10	UART4_RX	3.3 V
11	GND	GND

FC DEBUG · 10-PIN JST-SH

PIN	SIGNAL	VOLTAGE
1	3V3_FMU	3.3 V
2	USART3_TX	3.3 V
3	USART3_RX	3.3 V
4	FMU_SWDIO	3.3 V
5	FMU_SWCLK	3.3 V
6	SPI6_SCK	3.3 V
7	NFC_GPIO	3.3 V
8	PD15	3.3 V
9	FMU_NRST	3.3 V
10	GND	GND

GPS · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VDD_5V_HIPOWER	5 V
2	USART1_TX	3.3 V
3	USART1_RX	3.3 V
4	I2C1_SCL	3.3 V
5	I2C1_SDA	3.3 V
6	GND	GND

TELEM1 · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VDD_5V_HIPOWER	5 V
2	UART7_TX	3.3 V
3	UART7_RX	3.3 V
4	UART7_CTS	3.3 V
5	UART7_RTS	3.3 V
6	GND	GND

CAN · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VDD_5V_HIPOWER	5 V
2	CAN1_P	5 V
3	CAN1_N	5 V
4	GND	GND

RC · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VDD_5V_SBUS_RC	5 V
2	USART6_RX	3.3 V
3	USART6_TX	3.3 V
4	GND	GND

PI UART3 · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5.0V	5 V
2	UART3_TX	3.3 V
3	UART3_RX	3.3 V
4	UART3_CTS	3.3 V
5	UART3_RTS	3.3 V
6	GND	GND

PI GPIO · 6-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5.0V	5 V
2	GPIO21	3.3 V
3	GPIO22	3.3 V
4	GPIO23	3.3 V
5	GPIO24	3.3 V
6	GND	GND

PI CONSOLE · 6-PIN JST-SH

PIN	SIGNAL	VOLTAGE
1	3.3V_RPI	3.3 V
2	CONSOLE_TXD0	3.3 V
3	CONSOLE_RXD0	3.3 V
4	NC	—
5	NC	—
6	GND	GND

PI LED STRIP · 8-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5.0V	5 V
2	5.0V	5 V
3	GPIO12	3.3 V
4	GPIO16	3.3 V
5	GPIO17	3.3 V
6	GPIO20	3.3 V
7	GND	GND
8	GND	GND

PI I2C1 · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	5.0V	5 V
2	I2C1_SCL	3.3 V
3	I2C1_SDA	3.3 V
4	GND	GND

PI ETH · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	ETH_RD_N	—
2	ETH_RD_P	—
3	ETH_TD_N	—
4	ETH_TD_P	—

PI FAN · 4-PIN MOLEX KK 254

PIN	SIGNAL	VOLTAGE
1	GND	GND
2	5.0V	5 V
3	FAN_TACH_CON	5 V
4	FAN_PWM_Q*	5 V

PI USB (HUB PORT 2) · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VBUS2_FLT	5 V
2	USB2_N	3.3 V
3	USB2_P	3.3 V
4	GND	GND

PI USB (HUB PORT 4) · 4-PIN JST-GH

PIN	SIGNAL	VOLTAGE
1	VBUS4_FLT	5 V
2	USB4_N	3.3 V
3	USB4_P	3.3 V
4	GND	GND

ARK-DS-ARK-PI6X | Version 1.0, 2026-10-30 | Compiled from the ARK Compute Module FC Production schematic Rev 2.0 (11 Apr 2024), the ARK Pi6X Flow Rev 2.0 STEP model, the PX4 and ArduPilot board definitions, the MIC2026A-76A data sheet (DS20006608A), and ARK Electronics published documentation. Temperature ratings per component data sheets. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.